Name the force and its source

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify forces on a chosen object and name each interaction partner.

SUCCESS CRITERIA - I can...

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “weight” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

weight: the gravitational force on an object

normal force: a support force perpendicular to a contact surface

01 Warm-up: bring back an earlier idea

A cart travels west at 12 m/s and west at 4 m/s 3 s later. Is it speeding up?
Find acceleration size and direction.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$.
F_g is weight; m is mass. Use g = 10 N/kg for CPA. Name the source of every force.

Worked reasoning: A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

Your turn: Draw the book's force diagram.
Name the force on Earth paired with the book's weight.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Add and label force arrows.

Book resting on a level table; draw forces on the book only.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Which object gets the force?

A book rests on a level table.
Draw its two vertical forces and name who exerts each.

Sentence starter: The … in my model represents … . Its label shows … .

Add and label force arrows.

The dot represents the object. Chosen object: book

Marzano check - circle: 1 2 3

04 Mass and weight

A 2 kg bag hangs at rest from a vertical rope.
Use g = 10 N/kg.
Draw the forces on the bag, then find its weight and the rope tension.
Label each force with its source.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Chosen object: hanging bag.

Marzano check - circle: 1 2 3

05 After the throw

A ball is moving upward after it leaves your hand. Ignore air resistance.
Draw its force diagram and name the source of every force.
Explain why upward motion does not require an upward force.

Sentence starter: My claim is … . The diagram/data show …, so … .

Add and label force arrows.

Ball after release; velocity is upward.

Marzano check - circle: 1 2 3

Name the force and its source

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Continue down each column.

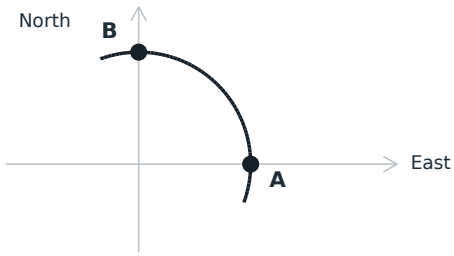
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A rover travels straight east at a steady 5 m/s. A second rover turns at the same steady speed. Which can have zero net force?
Explain using velocity.
Use the curve for the turning case; show how velocity changes.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .



Path sketch: runner moves from A toward B.

Marzano check - circle: ① ② ③

07 Exit check

A 3 kg lamp hangs still from one vertical cord.
Use g=10 N/kg.
Name both forces with source, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Add and label force arrows.

Hanging object at rest: label sources and magnitudes.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A hand presses down on the resting book. Add the new force and explain how the table's support must change.

Sentence starter: My claim is … . The diagram/data show …, so … .

Add and label force arrows.

Book pressed down by a hand: draw forces on the book.

Marzano check - circle: ① ② ③